

Introduction to Nutanix and AOS - Day 4

Cassandra

Cassandra stores and manages all of the cluster metadata in a distributed ring-like manner based upon a heavily modified Apache Cassandra. The Paxos algorithm is utilized to enforce strict consistency. This service runs on every node in the cluster.

The Cassandra is accessed via an interface called Medusa.

Components of Cassandra service

Cassandra database

Stores and manages all cluster metadata

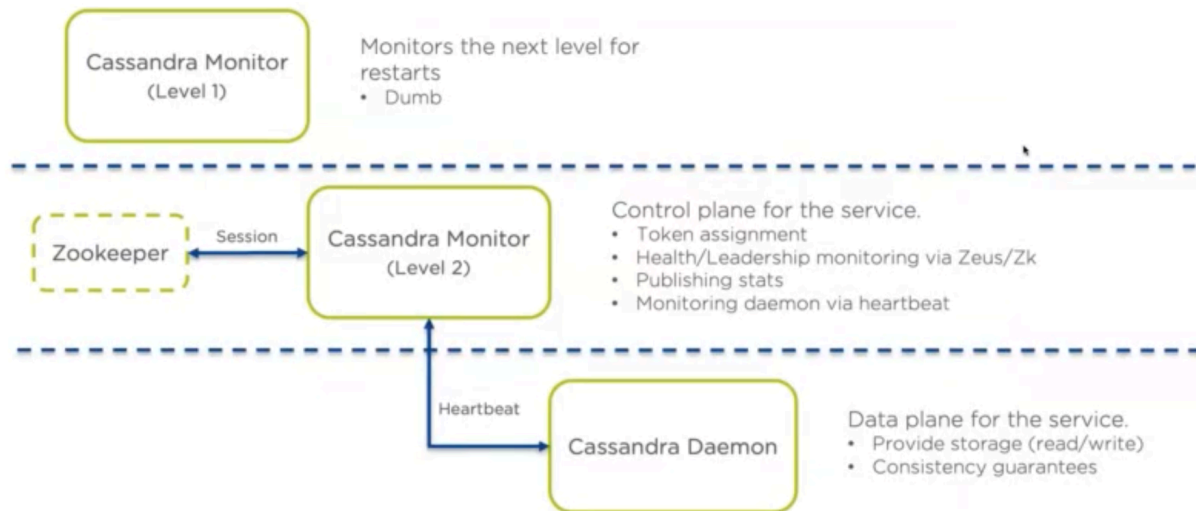
Medusa

The software interface used to access cassandra

Dynamic Ring Changer

Witness process that monitors cassandra status and manages removals/additions for the cassandra ring

Cassandra service architecture



In Cassandra, we have a three level hierarchy:

- At Lowest level, we have Cassandra daemon, that performs the actual storage/metadata operations. This is the data plane for Cassandra
- Next, we have a Cassandra monitor that acts as the control plane for service and performs various activities like token assignment, health monitoring in addition to monitoring the daemon for restarts
- Top level Cassandra monitor is used to monitor the control plane process

CVM States

Cassandra on a CVM can be in the following states:

Pre-Limbo

This is the state when the node is not a part of the metadata ring yet, but it is available and waiting to be added to the ring

Limbo

This is the state when the node is in the process of receiving copies of the metadata it owns as part of the token ownership from the neighbour nodes in the ring

Normal

This is the state when the node is healthy and actively participating in the metadata operations

Forwarding

This is a state in which node is not actively participating in the metadata operations and forwards any metadata requests from the local services (Stargate, Curator, Pithos, etc) to remote Cassandras. There are different types of events that trigger Cassandra to move into forwarding mode:

- CVM is marked for removal from the metadata ring due to reasons such as CVM down for more than 30 minutes or Cassandra services crashing 7 times in the last 30 minutes
- Metadata on the node needs repair due to local health warnings such as missing or corrupt files or disk failures
- CVM is marked degraded by the cluster. One of the corrective action is to put the Cassandra on the degraded node in a forwarding state
- CVM is down in a 3 node cluster

```
nutanix@NTNX-19FM6J120203-B-CVM:10.24.138.237:~$ nodetool -h 0 ring
```

Address	Status	State	Load	Owns	Token
kV000000kcXGKG3h0USc3aLGMpmCdq0H0ZLTEAquEiJYRvLko43S0fu8u1wo					
10.24.138.239	Up	Normal	30.21 MB	25.00%	00000000wp1CCoNLF7SDIXYbpy00nd1wTNxj3y6BBY911yRffDFBQnIWR46H
10.24.138.236	Up	Normal	27.84 MB	25.00%	FV000000Z8AMPGb7o0D4TKdAZtdQRLUffAmUJpWBxkj3600i7iRmosQwr4vS
10.24.138.237	Up	Normal	28.61 MB	25.00%	V00000008uTYSRXe531yeuU68EX0ut6KB44prN9Ug9ACLPiurAaaabDixlj
10.24.138.238	Up	Normal	31.42 MB	25.00%	kV000000kcXGKG3h0USc3aLGMpmCdq0H0ZLTEAquEiJYRvLko43S0fu8u1wo

Cassandra WAL - Write Ahead Log

- Scheduler VM WAL
- A strong consistency view of VM resource allocation during Acropolis leadership changes

Cassandra Logs

Level 1 Cassandra Monitor logs are found in : `~/data/logs/cassandra.out`

Level 2 Cassandra Monitor logs are found in: `~/data/logs/cassandra_monitor.INFO`

Cassandra Daemon logs can be found in: `~/data/logs/cassandra/system.log.`
`[INFO|WARN|ERROR]`

Dynamic Ring Changer logs

Tracks ring addition/removal operations, and will note any issues with this process.

Logs found in: `~/data/logs/dynamic_ring_changer.out[INFO|WARN|ERROR]`

You should only check this log if you think there is an issue with ring removal/additions

PithosDB

Port: 2016

- Persistent store for all of VM configs and parcels
- Host Usage
 - Acropolis peer on each host reports actual host usage(every 30 seconds).
 - Acropolis Scheduler on master listens to host usage change and adjust its resource scheduling

Scheduler

Give best priority to nodes that have both plenty of RAM and CPU that are unused.

Parcels

- Backed by Pithos
- Keep tracks of parcels allocated in the cluster.
- Parcel DB is not always used - most of the times, we always read it from WAL

All parcels list can be viewed using :

```

acli parcel.list
nutanix@NTNX-16SM65230111-A-CVM:10.136.106.126:~$ acli parce
l.list
Parcel name                                Parcel UUID
05c7fbd5-dbc8-40e4-9926-3e15ee819fce    05c7fbd5-dbc8-40e4-9926
-3e15ee819fce
06b14098-bd7d-4f46-8598-e5bb808bb3e0    06b14098-bd7d-4f46-8598
-e5bb808bb3e0
085a56c9-c0b1-4534-b92b-777752e80ef8    085a56c9-c0b1-4534-b92b
-777752e80ef8

```

To get details of an individual parcel:

```

nutanix@NTNX-16SM65230111-A-CVM:10.136.106.126:~$ acli parce
l.get 05c7fbd5-dbc8-40e4-9926-3e15ee819fce
{
  config {
    affinity_spec {
    }
    configured_memory_bytes: 4294967296
    cross_cluster_live_migrate_in_progress: False
    entity {
      entity_name: "Ajith-Windows"
      entity_type: "kVM"
      uuid: "0b812c3c-cb66-4eca-a405-52eca750db7b"
    }
    gpu_console: False
    ha_protected: True
    memory_bytes: 4294967296
    memory_overcommit: False
    num_cpus: 4
    num_vnics: 1
    use_segments: False
  }
}

```

```
logical_timestamp: 3
node_uuid: "ea6fa17e-5cd7-4853-b6ca-99f80463a48c"
owner_task_uuid: "639215d8-8b0c-4374-b1f1-98382eebb33b"
uuid: "05c7fbd5-dbc8-40e4-9926-3e15ee819fce"
}
```

Curator

- Curator is responsible for managing and distributing tasks throughout the cluster, including disk balancing, proactive scrubbing, and many more items.
- Curator is a background process which runs on all the nodes in a cluster
- Periodically scans metadata present in Cassandra
- It works in a master-slave architecture with the Master orchestrating all the jobs and scans
- Curator does not scan/read any actual data on the cluster

Curator uses MapReduce infrastructure to work on scanned data and generates tasks.

Chronos

- Chronos is responsible for taking the jobs and tasks resulting from a Curator scan and scheduling/throttling tasks among nodes
- Runs inside Curator master
- Receives background tasks and distributes them to all Stargates in the cluster
- Rate limits for 10 tasks per Stargate
- Takes care of tasks priorities
- Dedups the tasks
- Chronos runs on every node and is controlled by an elected Chronos Master that is responsible for the task and job delegation and runs on the same node as the Curator Master

Types of Curator scans

- Full Scan - Scans all the Metadata Maps in the cluster
- Partial Scan – Scans a subset of Metadata Maps
- Selective Scan – Scans a subset of Metadata Maps based on the reason why the scan was triggered

i.vDiskSevering

ii.ILM (Information Lifecycle Management)

iii.NFS Map

iv.Node/Disk Failure

Curator Scan Schedules

- Periodic
 - 1 Hour for Partial Scan
 - 6 Hours for Full Scan
- Dynamic
 - Triggered by a change in the cluster
 - Disk/Node Failure
 - High Tier Usage
 - Free Space
 - Container RF update

Scan Intervals

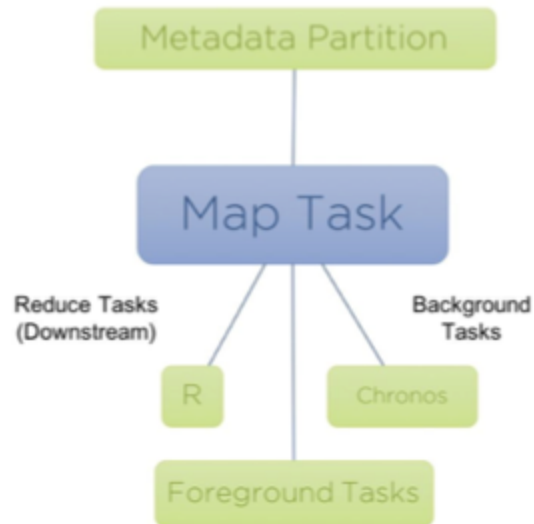
Each Scan consists of one or more MapReduce jobs.

- Jobs per Scan

i.1 for Selective Scan

ii.1 for Partial Scan

iii.4 for Full Scan



Map Reduce Tasks

Key value pair is given to Reduce

Nodes are assigned with partition numbers given by Curator master

Nodes get the key value pairs based on partition number and perform the reduce tasks



DR - Cerebro

- DR, Backup, Snapshot - Meaning and Differences
- RTO and RPO
- DR Constructs
- Snapshot Workflow
- MetaOp
- Replication Stages
- AsyncDR
- NearSync DR
- Troubleshooting and Lab Demo

DR, Backup, Snapshot - Meaning and Differences

Disaster Recovery

Disaster Recovery solutions ensure that a datacenter can be protected from the effects of a natural or man-made catastrophe.

Backup

Backup refers to the process of making copies of data to use in case the original data is lost/destroyed.

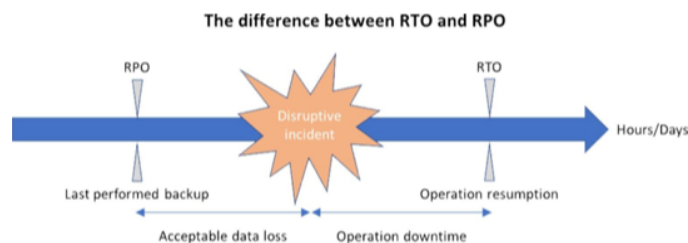
Snapshot

Snapshots are a point in time copy of data to which you can roll-back at any time.



Backups are full copies of data.

Snapshots are pointers to the data.



Recovery Point Objective (RPO)

A time interval that refers to the acceptable data loss if a failure occurs.

Recovery Time Objective (RTO)

The time period from failure event to the restored service.

Protection Domain

A protection domain is a defined group of entities (VM/VG/NFS files) that must be protected.

Consistency Group

A consistency group is a set of VMs or files that will be backed up as snapshot at an exact moment in time.



Schedules

Schedules establish the frequency of snapshots and replication within the PD.

Retention Policy

Each schedule has a retention policy that determines the number of local and remote snapshots that will be retained.

Remote Site

Administrators can define remote sites so that snapshots can be stored away from the site where the VM is running.

Remote Site Config.

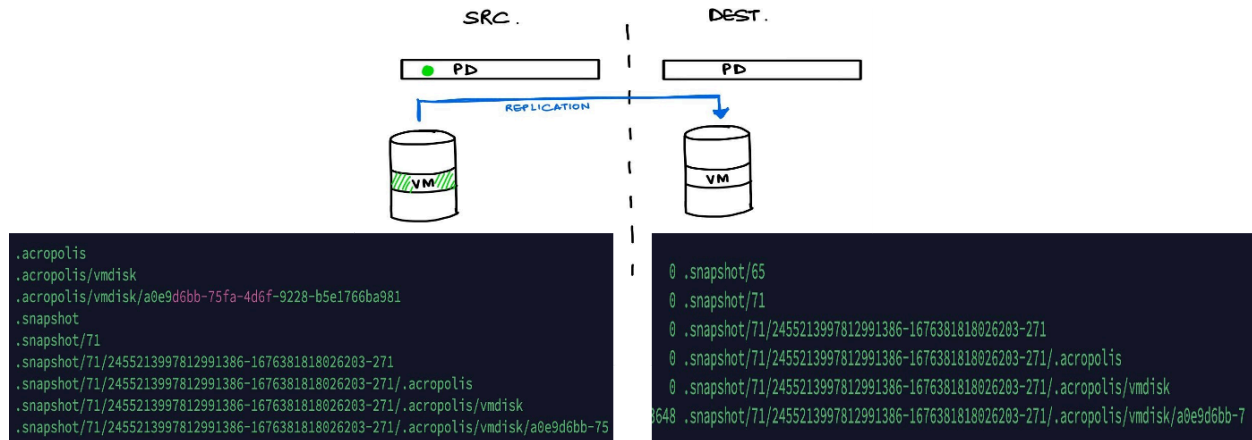
Remote site (destination cluster) configuration is stored in Zeus. Defined by -

- Mapping of containers
- Mapping of network
- RS used for DR or backup
- Configurable Data transfer rate, throttling yes/no.
- Compression on Wire

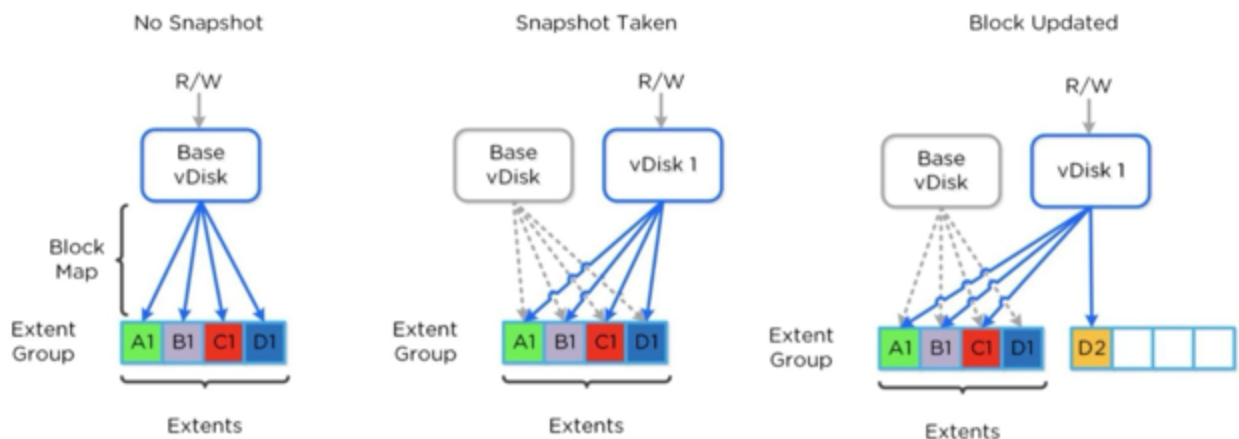
I20230214 14:12:01.699286Z 4730 add_remote_op.cc:469] [Remote Name: Dizzy, opid: 451]Added remote site of type kLegacy to the zeus config

Active Protection Domain

Inactive Protection Domain



Snapshot Workflow



Async DR

- Asynchronous Replication of Protection Domains
- RPO as low as 60 minutes
- Spans across containers

- Can replicate to many clusters

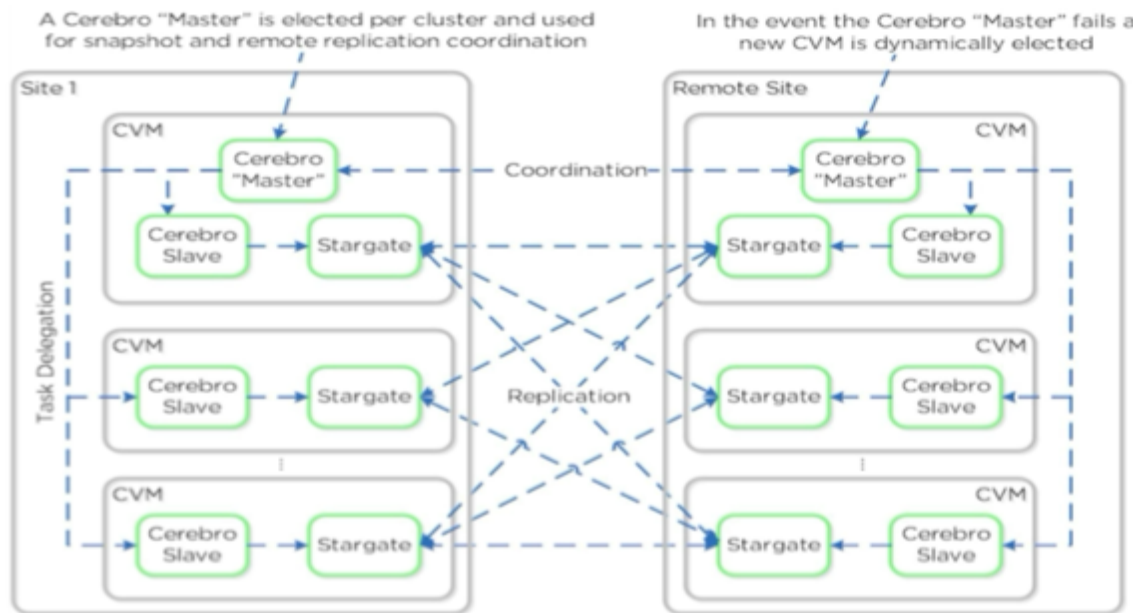
The first ever snapshot will be replicated in FULL.

The last successfully replicated snapshot to a remote site is retained as REFERENCE snapshot.

Subsequent snapshots consider DELTAS to the reference snapshot.

VDiskDiffOp uses current reference snapshot and snapshot being replicated to calculate the vblocks that changed since Reference snapshot.

Cerebro Workflow



Replication Stages

Replication is the process of asynchronously copying snapshots from one cluster to one or more remote sites.

Stages -

Local Cerebro master transfers the Protection Domain meta-data

Remote Cerebro master creates the Protection Domain if it doesn't exist

Local Cerebro master transfers the Snapshot Metadata (file paths)

Local Cerebro master distributes work of replicating files to Cerebro slaves on local cluster (unit of work is a file)

Local Cerebro slave computes the differences in vblocks between the current snapshot and the previous snapshot

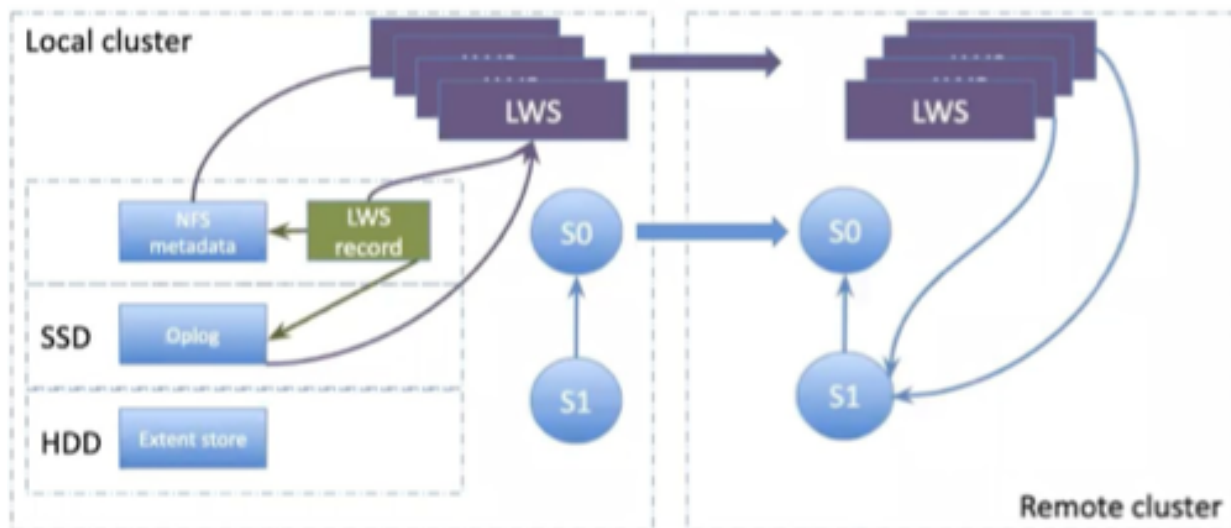
Local Cerebro slave communicates with the local Stargate to setup replication of first the file paths, followed by replication of the vblocks.

Local stargate (vdisk_controller) coordinates the replication of vblocks to the remote stargate vdisk_controller.

Cerebro master queries cerebro slaves for completion of the replication task.

Nearsync DR

- RPO between 1 - 15 mins.
- Makes use of light-weight snapshots (LWS) for quick replication.
- NearSync automatically transitions out if it can't keep up with schedule.
- All writes land on SSD - never on the HDD.



Labs:

simulating disk failure

simulating CVM failure

Simulating node failure - HA